



Give Bit the Steps

Write the arrows to reach the star

Name: _____

Date: _____

★ I can write the arrow steps that get Bit to the star.

You make the plan

Now YOU give Bit the steps. Look at where Bit is and where the star is. Draw one arrow for each box Bit must move. ➡ goes right, ⬅ goes left.

Watch Bit — box 1 to the star on box 3



Bit must move 2 boxes right. The steps are ➡ ➡.

1 Box 1 to the star on box 4

Bit is on box 1. The star is on box 4. Draw the arrows Bit needs (one per box).

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2 Box 2 to the star on box 3

Bit is on box 2. The star is on box 3. Draw the arrows Bit needs.

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3 You try — go left

Bit is on box 4. The star is on box 2. Draw the arrows Bit needs (this time the star is to the left).

Two empty rectangular boxes with rounded corners and thin black borders, intended for drawing arrows. They are positioned side-by-side.

Big idea

When you write the steps that solve a problem, you have made an algorithm. That is exactly what coders do!



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Quick facilitation notes & answer key

What this worksheet teaches

Children draw a short set of arrows that moves Bit to the star — choosing the direction and how many steps. Writing the steps that solve a problem is making a simple plan (an algorithm). No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil or crayon. You read each prompt aloud. Optional: tape a row of numbered boxes (and a star) on the floor so a child can walk the steps.

How to lead it (about 15 minutes)

1. Show them (you do). Box 1 to the star on box 3: two right arrows.
2. Do one together. Exercise 1 — box 1 to box 4 needs three right arrows.
3. Let them try. Children do Exercise 2 (one right) and Exercise 3 (two left — the star is to the left).

Answer key (these are the verified correct answers)

Exercise 1 — box 1 → box 4: three right arrows. Exercise 2 — box 2 → box 3: one right arrow.

Exercise 3 — box 4 → box 2: two left arrows.

If children get stuck — and ways to adjust

Watch for: right number of steps but wrong direction. Check which way the star is first.

Make it easier: count the boxes between Bit and the star, then draw that many arrows.

Make it harder: ask for a plan from box 1 to box 5 (four right arrows).

Make it active: a child walks their drawn arrows on the floor boxes to test the plan.

Ontario Kindergarten link (official wording)

Problem Solving and Innovating — children create a sequence of directions to reach a goal.

In plain terms: writing the steps that solve a problem — making a simple plan.

No reading is needed from the child — you read each prompt aloud; the child answers by circling, numbering, or drawing arrows.

You'll know it worked when

The child draws arrows in the right direction and the right number to reach the star.